

Mengyi Chen

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EDUCATION

National University of Singapore

Ph.D. in Mathematics (Expected Aug. 2027)

Singapore

Aug. 2023 – Present

- Supervisor: Qianxiao Li
- Research Interests: Machine learning for modeling the macroscopic dynamics of complex systems; multiscale modeling; AI for materials

Columbia University

Visiting Student, School of Engineering

New York, USA

Jan. 2022 – May 2022

University of Chinese Academy of Sciences

B.Sc. in Mathematics and Applied Mathematics

Beijing, China

Sep. 2019 – Jun. 2023

PUBLICATIONS

- [1] **Mengyi Chen**, Peichen Zhong, Zihan Zhang, Qianxiao Li. “Learning Ab Initio Phase-Field Models.” *arXiv preprint* arXiv:2610.01432, 2026.
- [2] Zihan Zhang*, **Mengyi Chen***, Qianxiao Li, Peichen Zhong. “Interfacial Melting as a Thermodynamic Indicator of Solid-State Synthesizability.” *arXiv preprint* arXiv:2606.22885, 2026.
* *These authors contributed equally.*
- [3] Zhichao Han, **Mengyi Chen**, Qianxiao Li. “Learning Permutation-Invariant Macroscopic Dynamics.” *International Conference on Machine Learning (ICML)*, 2026 (accepted).
- [4] **Mengyi Chen**, Pengru Huang, Kostya S. Novoselov, Qianxiao Li. “Scalable Learning of Macroscopic Stochastic Dynamics.” *Physical Review Materials*, 10(3):033805, 2026.
- [5] **Mengyi Chen**, Qianxiao Li. “Learning Macroscopic Dynamics from Partial Microscopic Observations.” *Advances in Neural Information Processing Systems (NeurIPS)*, 37:48996–49021, 2024.
- [6] Sheng Gong, Yumin Zhang, Zhenliang Mu, Zhichen Pu, Hongyi Wang, Xu Han, Zhiao Yu, **Mengyi Chen**, Tianze Zheng, Zhi Wang, Lifei Chen, Zhenze Yang, Xiaojie Wu, Shaochen Shi, Weihao Gao, Wen Yan, Liang Xiang. “A Predictive Machine Learning Force-Field Framework for Liquid Electrolyte Development.” *Nature Machine Intelligence*, 7(4):543–552, 2025.

WORK EXPERIENCE

ByteDance AML Research Intern

Dec. 2022 – Jul. 2023

Beijing, China

- Contributed to developing equivariant graph neural network models for machine learning force fields in the ByteDance AI Molecular Simulation Booster (BAMBOO) project

HONORS & AWARDS

NUS Faculty of Science PhD Conference Award	2026
NUS Graduate Tutor Merit Award (Semester 2, AY2025/26)	2026
NeurIPS 2024 Travel Award	2024
NUS Faculty of Science Dean’s Graduate Fellowship	2023
UCAS Overseas Graduate Studies Fellowship (Top 5%)	2023
UCAS Second-Class Scholarship	2022

PROFESSIONAL SERVICE

Co-organizer, <i>ICLR 2025 Workshop on Machine Learning Multiscale Processes (MLMP)</i> , Singapore	2025
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SKILLS & INTERESTS

Languages: Chinese (native), English (fluent)

Interests: Rock climbing, violin